

VCU Project Documentation

1. What It Does

Controls electric vehicle with 4 states:

- INIT: Safe start, everything off
- STARTIN: Button + brake for 2s + BMS OK
- BUZZIN: Buzzer on for 2s
- DRIVE: Read pedals, control motor, check faults

2. Safety Features

- APPS fault: Sensors differ >10% for >100ms → stop motor, go to INIT
- Brake light: Turns on when brake >10%
- Motor: Only runs in DRIVE state, zero torque otherwise

3. CAN Communication

3 CAN buses:

1. Motor (0x201): Send torque commands
2. BMS: Listen for OK signal (0x50 in byte 6)
3. Debug: Send status every 100ms

4. How It Works

loop() every 10ms:

1. Read buttons/pedals
2. Check current state
3. Do state actions
4. Update outputs

5. Key Settings (change in code)

```
BRAKE_THRESH = 102      // 10% of 1023
STARTIN_HOLD_TIME_MS = 2000 // 2 seconds
FAULT_HOLD_TIME_MS = 100  // 100ms fault time
FLIP_MOTOR_DIRECTION = false // Reverse motor if true
```

6. Testing

- Serial monitor: 9600 baud for state info
- CAN debug: Use SavvyCAN with .dbc file
- Hardware: LEDs/buzzer show state

7. Maintenance

- Change pins: Update #define at top of file
- Add features: Modify DRIVE state section
- Debug: Check APPS scaling (×50/33 formula)